OroTan-MX

Carbomer–Hypromellose Hybrid Mucoadhesive Oral Gel Platform

A Standardized Phytopharmaceutical Drug Delivery System for Intraoral Mucosal Wound Management

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BUSINESS DEVELOPMENT & LICENSING PROPOSAL

Prepared for

Dr. Reddy's Laboratories Ltd.
New Product Development / Business Development & Licensing

Prepared by

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Executive Summary

OroTan-MX is a dual-polymer mucoadhesive oral gel platform designed for the localized management of intraoral mucosal lesions, including recurrent aphthous ulcers, traumatic mucosal injuries, and chemotherapy- or radiotherapy-induced mucositis. The platform combines a synthetic anaesthetic active pharmaceutical ingredient (Lignocaine Hydrochloride) with a standardized matrix of purified, plant-derived polyphenolic and tannin fractions, delivered through a stabilized carbomer–hypromellose hydrogel network.

The core technical differentiator of the platform is the use of Hypromellose K100M as a protective colloid that sterically shields the carbomer network from polyphenol-induced precipitation — a limitation that has historically constrained the use of high-tannin botanical actives in single-polymer hydrogel systems. This allows a 3.0% w/w standardized polyphenol load to be carried in a stable, clear gel suitable for topical intraoral application.

This proposal is submitted to Dr. Reddy’s Laboratories Ltd. for scientific evaluation by the New Product Development and Business Development & Licensing teams, with a view to establishing a collaborative pathway — through licensing, co-development, or contract manufacturing — for the advancement of OroTan-MX toward formal analytical validation, regulatory filing, and commercialization.

1. Unmet Clinical Need & Market Rationale

Intraoral mucosal lesions — encompassing recurrent aphthous stomatitis, traumatic ulcers arising from orthodontic appliances or dentures, and mucositis secondary to chemotherapy or radiotherapy — represent a persistent and underserved category within topical oral care. Currently marketed topical products typically address only a single dimension of the clinical need: anaesthesia, astringency, or barrier protection, but rarely all three within a single, stable formulation.

A recognized formulation constraint underlies this gap. Polyphenolic and tannin-rich botanical actives, which provide astringent, protein-cross-linking benefits for wound-bed protection, are largely incompatible with conventional single-polymer carbomer hydrogels, as they tend to precipitate the polymer network on contact. This has historically limited the practical use of high-potency botanical astringents within stable topical gel formats.

OroTan-MX is positioned to address this gap directly — combining rapid local anaesthesia, astringent wound-bed protection, and extended mucosal contact time within a single, mucoadhesive, allopathic phytopharmaceutical platform suitable for evaluation under conventional drug regulatory frameworks.

2. Technology Platform Overview

OroTan-MX is built on a flash-neutralization, dual-polymer hydrogel architecture combining Carbomer 974P (0.80 g/100 g) as the primary structural network with Hypromellose K100M (1.00 g/100 g) as a protective colloid and long-acting bioadhesive enhancer. Table 1 summarizes the principal architectural elements of the platform and their functional role.

Table 1. Platform Architecture — Functional Overview

Component	Functional Role
Carbomer 974P	High-yield structural hydrogel network anchor.
Hypromellose K100M	Protective colloid barrier; shields the carbomer network from tannin-induced precipitation and extends mucosal residence time. [1]
Flash-neutralization (10% NaOH, target pH 5.5–6.0)	Triggers controlled gelation while maintaining a pH window compatible with topical mucosal delivery and Lignocaine HCl ionization balance.
Buffer & preservative system (citrate / EDTA / benzoate / sorbate)	Maintains electrolyte stability, pH control, and microbiological preservation across shelf life.

Hypromellose's function as a protective colloid that sterically stabilizes a co-dispersed phase against precipitation is a documented property of the polymer in other systems. [1] The application of this principle to stabilize a high-tannin carbomer hydrogel specifically is the platform's own technical innovation and has not been independently published; formal characterization of this mechanism within the OroTan-MX matrix is proposed as an analytical workstream in Section 7.

3. Active Ingredient Matrix & Mechanistic Rationale

Table 2 summarizes the active pharmaceutical ingredient (API) matrix, standardization targets, and the proposed mechanistic rationale for each component, consistent with the standardized botanical nomenclature conventions used throughout the technical dossier (Annexure A).

Table 2. API Matrix — Standardization & Mechanistic Rationale

API Component	Standardization Target	Qty / 100 g	Mechanistic Rationale
Lignocaine Hydrochloride (synthetic API)	≥ 99.0% lidocaine HCl (HPLC)	2.000 g	Voltage-gated sodium channel blocker; provides rapid, reliable local anaesthesia at the ulcer wound bed. [2]
Punica granatum pericarp fraction	≥ 10% punicalagins (HPLC)	3.000 g	Tannin-mediated protein cross-linking for astringent action; forms a protective pellicle over the lesion bed. [3]
Quercus infectoria gall fraction	≥ 50% gallotannins (HPLC)	0.0413 g	Astringent-mediated tissue tightening; rapid-onset, complementary tissue-binding action. [4]
Polyphenolic tri-fruit complex (Phyllanthus emblica / tri-fruit fractions)	≥ 25% total tannins (HPLC)	0.1239 g	Broad-spectrum antioxidant and anti-inflammatory adjunct, as reported in the phytopharmacological literature. [5]
Clove (Syzygium aromaticum) essential oil fraction	≥ 85% eugenol (GC-MS)	0.0413 g	Secondary sensory analgesic / counter-irritant action, complementary to Lignocaine HCl. [6]
Purified camphor fraction	Aromatic assay purity	0.0080 g	Activation of TRPM8 cold-thermoreceptors, producing a cooling counter-irritant sensation. [7]
Piperine-gingerol complex	≥ 95% piperine fraction (HPLC)	0.0618 g	Permeation-supportive adjunct; mechanistic rationale for mucosal delivery enhancement detailed in Section 4.

Every active constituent in the OroTan-MX platform is supplied as a purified, standardized plant-derived chemical fraction, characterized against an analytical specification (HPLC or GC-MS). No raw botanical material is used at any stage of manufacture, consistent with the platform’s positioning as an allopathic phytopharmaceutical drug-delivery system rather than a traditional herbal formulation.

Sourcing for each fraction is intended to draw on GMP-compliant vendors whose raw APIs already carry validated HPLC purity profiles, low residual solvent specifications, and certified heavy metal limits in accordance with ICH Q3D guidelines on elemental impurities, reducing the incremental analytical burden required to bring each component into a regulatory-grade specification package.

4. Scientific Rationale: Piperine-Gingerol Complex as a Mucosal Permeation Enhancer

4.1 Bioenhancers: Definition and Background

Bioenhancers are agents that are not independently therapeutic but, when co-administered with an active drug, potentiate its pharmacological effect — typically by improving absorption, slowing metabolism, or both. [8] Piperine, the principal pungent alkaloid of *Piper nigrum* and related *Piper* species, was scientifically validated as one of the first recognized bioavailability enhancers by Indian researchers at the Regional Research Laboratory, Jammu, in 1979, following earlier empirical observations of enhanced therapeutic effect when long pepper was co-administered with other botanical actives. [8]

4.2 Proposed Mechanisms of Action

Piperine's bioenhancing activity has been attributed to several overlapping mechanisms that are relevant to the OroTan-MX platform: modulation of cell membrane lipid dynamics, which increases tissue permeability at the site of absorption; inhibition of drug-metabolizing enzymes, including cytochrome P450 3A4 (CYP3A4), UDP-glucuronyl transferase, and aryl hydrocarbon hydroxylase; inhibition of P-glycoprotein-mediated efflux transport, which reduces clearance of the co-administered active from the absorbing tissue; and localized vasodilation, which increases blood flow at the site of absorption. [8] These mechanisms collectively form the published basis for piperine's classification as a bioavailability enhancer rather than as an independently therapeutic agent.

4.3 Quantitative Evidence from Established Drug Systems

The bioenhancing effect of piperine has been quantitatively demonstrated across a range of co-administered drugs and nutrients in published literature. Concomitant administration of piperine has been reported to increase the bioavailability of curcumin by approximately 2,000%, through inhibition of hepatic and intestinal glucuronidation. [8] In patients with pulmonary tuberculosis, piperine co-administration increased the bioavailability of rifampicin by approximately 60%, supporting a reduction in the standard rifampicin dose from 450 mg to 200 mg without loss of therapeutic effect. [8] Comparable enhancement has been reported for vasicine (greater than 200%) and sparteine (greater than 100%), among other co-administered actives. [8]

4.4 Relevance to the OroTan-MX Platform

Within OroTan-MX, the piperine-gingerol complex is included at a calibrated, sub-pungency dose (0.0618 g/100 g, ≥ 95% piperine fraction) as a permeation-supportive adjunct rather than as a primary therapeutic agent. The biochemical mechanisms underlying piperine's bioavailability-enhancing effects—including modulation of membrane lipid bilayers, localized vasodilatory activity, and inhibition of selected metabolic enzymes and efflux transporters—

are biologically relevant to oral mucosal tissue. The membrane-modulating and localized vasodilatory mechanisms described above are biologically plausible in the context of oral mucosal tissue and are consistent with the platform's overall objective of improving the residence time and local bioavailability of the co-formulated actives at the lesion site.

Consistent with standard pharmaceutical development practice, the magnitude of permeation enhancement attributable to this adjunct within the specific OroTan-MX matrix has not yet been experimentally confirmed and is proposed as a defined work-stream for the joint feasibility phase outlined in Section 9.

5. Formulation & Manufacturing Readiness

OroTan-MX is manufactured via a validated flash-neutralization process structured across six discrete phases (A through F):

- Phase A: Aqueous-phase solubilization of buffering agents, preservatives, polyol, sweetener, and Lignocaine Hydrochloride at 65°C under sealed conditions to prevent micro-evaporation.
- Phase B: Dry-blending and high-shear dispersion (1,500 rpm) of Carbomer 974P and Hypromellose K100M into the Phase A vortex, followed by a minimum four-hour hydration period.
- Phase C: Integration of the pre-dissolved tannin extract fractions into the hydrated polymer slurry.
- Phase D/E: Incorporation of the piperine-gingerol and aromatic phytochemical fractions, followed by polysorbate 80–stabilized micellar dispersion of the lipophilic clove and camphor fractions.
- Phase F: Controlled, drop-wise addition of 10% sodium hydroxide solution under continuous digital pH monitoring to reach the target gelation threshold of pH 5.5–6.0, followed by final weight adjustment with purified water.

This process has been designed for straightforward scale-up using standard pharmaceutical compounding equipment — borosilicate-jacketed vessels, high-shear and low-speed paddle mixers, and in-line pH monitoring — and is amenable to technology transfer review by Dr. Reddy's Laboratories' formulation development team.

6. Technical Risk Mitigation Matrix

Table 3. Technical Risk Factors & Engineering Mitigation

Technical Risk Factor	Mechanistic Basis	Engineering Mitigation
Tannin-induced polymer complexation	High polyphenol concentrations precipitate conventional single-polymer carbomer hydrogels.	Hypromellose K100M integrated as a protective steric/colloid barrier, preventing direct tannin–carbomer interaction. [1]

Technical Risk Factor	Mechanistic Basis	Engineering Mitigation
High formulation osmolality / mucosal stinging	Cumulative ionic solute and polyol load can osmotically draw fluid from open wounds.	Xylitol restricted to 3.0% w/w; low-mass sucralose used for taste-masking to lower solute particle load.
Lignocaine HCl ionization balance	Lignocaine HCl efficacy is pH-sensitive (pKa ≈ 7.9).	Citrate buffer system maintains a tight pH 5.5–6.0 post-neutralization window aligned to topical delivery requirements.
Mucosal permeability & sensory tolerability	Oral mucosal epithelium presents a significant absorption barrier; natural permeation enhancers can introduce pungency.	Piperine-gingerol complex dosed at a calibrated, sub-pungency level; mechanistic rationale supported by published literature (Section 4), with experimental confirmation proposed for the feasibility phase.

7. Regulatory Classification & Proposed Pathway

The inclusion of Lignocaine Hydrochloride 2% w/w as a synthetic, pharmacopeial-grade active pharmaceutical ingredient supports classification of OroTan-MX as a conventional drug product, rather than a traditional herbal or Ayurvedic formulation, notwithstanding the standardized botanical-derived polyphenolic fractions included in the API matrix. All botanical-derived constituents are specified and released against analytical standards (HPLC or GC-MS) rather than as raw plant material, consistent with regulatory expectations for phytopharmaceutical drug products.

The following analytical and validation milestones are proposed for the next development phase, and are put forward as candidate work-streams for joint execution with Dr. Reddy's Laboratories' analytical and regulatory affairs functions:

- Formal rheological characterization, including plastic yield value evaluation.
- In-vitro mucoadhesion profiling against porcine buccal mucosa.
- In-vitro permeation studies of Lignocaine Hydrochloride across artificial membranes.
- Long-term stability profiling under ICH-aligned climatic zone conditions.
- Vendor and raw-material qualification audits, confirming GMP compliance and validated certificates of analysis — HPLC purity profiles, residual solvent limits, and ICH Q3D heavy metal/elemental impurity limits — for each sourced API fraction.
- Experimental confirmation of the piperine-gingerol permeation-enhancement contribution within the specific OroTan-MX matrix.

8. Proposed Engagement Models

This proposal is structured to accommodate Dr. Reddy's Laboratories' preferred mode of engagement. Three indicative models are outlined below; a combination or staged progression across these models may also be considered.

Table 4. Indicative Engagement Models

Engagement Model	Description	Indicative Scope
Licensing	Transfer of technology rights for OroTan-MX to Dr. Reddy's Laboratories for further development, regulatory filing, and commercialization.	Upfront and/or milestone-based licensing fee; royalty structure on net sales; inventor retains a defined consultative role through the validation phase.
Co-Development	Joint progression of the analytical validation work-streams outlined in Section 7, with shared technical resourcing and cost participation.	Joint feasibility and pilot-batch programme; shared IP and commercialization terms to be defined under a co-development agreement.
Contract Manufacturing	Dr. Reddy's Laboratories engaged as manufacturing partner following independent validation, with the inventor / originating party retaining commercial and regulatory ownership.	Technology transfer, scale-up batch manufacturing, and quality release under a manufacturing services agreement.

9. Next Steps

1. Execution of a mutually agreed Non-Disclosure Agreement to enable detailed technical review by Dr. Reddy's Laboratories' R&D and analytical teams.
2. Joint review of the full technical dossier (Annexure A) by the nominated scientific evaluation team.
3. Scoping discussion to confirm the preferred engagement model (Section 8) and define the joint feasibility work plan.
4. Initiation of pilot-batch manufacture and the analytical validation work-streams outlined in Section 7.
5. Joint go/no-go review ahead of formal regulatory filing and scale-up.

We welcome the opportunity to present this platform in further technical detail to Dr. Reddy's Laboratories' evaluation team and to address any scientific or commercial queries arising from this proposal.

For further information or to schedule a technical discussion, please contact:

[Name]
[Designation / Organization]
[Email] | [Phone]

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9. OroTan-MX Technical Dossier — Carbomer-Hypromellose Hybrid Mucoadhesive Oral Gel Platform, (Bioenhancer Rationale Updated), June 2026. [Annexure A — Confidential, provided under separate cover.]

Annexure A

The complete OroTan-MX Technical Dossier containing the full API matrix, excipient specification, master batch composition, validated manufacturing protocol, and proposed analytical validation plan referenced throughout this proposal — accompanies this document as a companion technical annexure.